

Chance Krenzer

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EDUCATION

Case Western Reserve University, Cleveland, OH **Expected May 2029**

B.S. Mechanical Engineering and B.S. Aerospace Engineering (double major)

GPA 3.80/4.00 | Dean's High Honors List | National Merit Scholar | NCAA Varsity Cross Country and Track & Field (800m)

Coursework: Mechanical Manufacturing, Strength of Materials, Dynamics, Chemistry of Materials, Thermodynamics

ENGINEERING DESIGN TEAMS

CWRU High-Speed Quadcopter | *Mechanical Design Lead*, six-person team targeting 500 km/h Dec 2025 to Present

- Lead the mechanical design and handle all electronics for a six-person team building a quadcopter to fly a 500 km/h level pass.
- Designed the 16-part printed airframe around heat-set inserts and screws, minimizing unique screw types to simplify assembly and weighing ease of assembly against ease of manufacturing.
- Printed 4 arm and motor-mount prototypes to fix wire routing, rotor rub and drag from exposed screw holes (wider wire slot, rotor recess, screw-on fairing cap), so the full shell printed with zero changes.
- Selected carbon-fiber-reinforced nylon over PETG for the printed shell through comparative static FEA, meeting a safety factor above 3 at 5% lower mass.
- Used 1 kHz pre- and post-filter IMU sampling to identify a 157 Hz motor-rotation vibration that accounted for 97% of roll-axis control-output variance and was linked to chronic motor overheating. Specified a harmonic notch filter to reject it.
- Sized a 2.6 kg laminar-body airframe (470 mm long, 93 mm diameter) with a minimum-drag geometry search after showing the earlier 88 mm body could only fit the battery if it grew to 652 mm.

CWRU VTOL Team | *Mechanical Team Specialist*, 3rd of 20 teams at 2026 Championship Aug 2025 to Present

- Redesigned 3D-printed landing gear that loosened and detached in flight testing into waterjet-cut aluminum legs, milled and tapped for bolted hardware, cutting landing-gear mass 10% while increasing stiffness.
- Machined 4 aluminum motor mounts and 3 landing-gear attachments on the CNC waterjet, mill and belt sander.
- Designed a bonding jig that held all 3 landing legs in alignment during epoxy cure so they matched in angle and position.
- Co-designed and fabricated a jig to bond 4 machined aluminum motor mounts to carbon-fiber arms for the X8 configuration.

PROFESSIONAL EXPERIENCE

Dronefly, Los Angeles, CA | *Drone Repair & Operations Intern (Part-time)* May to Aug 2026

- Diagnosed and repaired commercial drones returned under warranty claims.
- Rebuilt the warranty and repair records system to give end-to-end traceability for every returned unit.
- Packaged and shipped 100+ drones and accessories to customers.

Los Angeles Department of Water and Power | *DERMS Fellow (Part-time)* May to Jul 2026

- Synthesized technical findings with a cross-functional team into a presentation for utility stakeholders.
- Analyzed gaps from growing DER adoption and designed DERMS approaches to help grid operators monitor and manage them.

PROJECTS

ESP32 Quadcopter | *Personal project* Dec 2025 to Feb 2026

- Designed the frame from a blank SolidWorks part, checked it with FEA and 3D-printed it in PETG, tuning print settings for arm strength at minimum mass.
- Built an N-channel MOSFET motor driver with flyback diodes and gate resistors, hand-soldering the surface-mount MOSFETs.
- Traced voltage sag seen only with all four motors running to an insufficient battery C-rating and fixed it with a 25C pack.

Steel Hammer | *Mechanical Manufacturing (EMAE 160), CWRU* Spring 2026

- Invited by the course professor to return as a teaching assistant for the manufacturing lab.
- Machined a cold-rolled steel hammer on the manual lathe and mill, holding handle diameter and head dimensions within 0.010 in. and passing go/no-go gauge checks.
- Worked with engineering drawings and GD&T callouts as part of the course.

EMG-Controlled Robotic Hand | *CWRU Hackathon, 4th of 12 teams* Nov 2025

- Built an EMG-controlled robotic hand with an electrical engineering teammate in a two-day hackathon and placed 4th of 12 teams.
- Designed hinge-jointed fingers in SolidWorks and merged a teammate's palm from another CAD tool via a neutral format.

TECHNICAL SKILLS

CAD & Drawings: SolidWorks (CSWA, PDM), parametric modeling, GD&T drawings (ASME Y14.5), DFM/DFA

Analysis: FEA (static, modal), SolidWorks Simulation, CFD (Flow Sim, Ansys Fluent), MATLAB

Manufacturing: 3D printing (FDM), CNC waterjet, mill, lathe, Mastercam, jigs and fixtures, adhesive bonding

Test & Integration: flight testing, ArduPilot log analysis, vibration analysis, PID tuning, root cause analysis, soldering

Certifications: Certified SolidWorks Associate (CSWA), OSHA 10-Hour Safety Training